

CET138 Bootstrap 5: The CSS Framework

Introduction

Bootstrap is the world's most popular front-end framework, originally developed by Twitter. It is a collection of pre-built HTML and CSS components, JavaScript plugins, and utility classes that dramatically speed up web development.

Bootstrap provides:

- 12-Column Grid System: A responsive layout system dividing rows into 12 equal columns.
- Pre-built Components: Ready-to-use cards, buttons, forms, navbars, alerts, modals, and carousels.
- Responsive Classes: Breakpoint-based utilities (col-md-6, d-flex) for mobile, tablet, and desktop.
- Utility Classes: Single-purpose classes for spacing, alignment, and display properties.
- JavaScript Plugins: Interactive components like dropdowns, tooltips, and modals.

Code Snippet Grid System & Navbar

```
<!-- Bootstrap Grid System Example -->
<div class="row g-4">
  <div class="col-sm-6 col-lg-3">
    <div class="concept-card">
      <h5>Elements and Tags</h5>
      <p>HTML is written using tags wrapped in angle brackets.</p>
    </div>
  </div>
  <div class="col-sm-6 col-lg-3">
    <div class="concept-card">
      <h5>Attributes</h5>
      <p>Attributes provide additional information about elements.</p>
    </div>
  </div>
</div>

<!-- Bootstrap Utility Classes -->
<nav class="navbar navbar-expand-lg fixed-top">
  <div class="container">
    <a class="navbar-brand" href="index.html">Rajesh Karki</a>
    <div class="d-flex align-items-center gap-2">
      <button class="btn btn-outline-primary">Login</button>
    </div>
  </div>
</nav>
```

Explanation of Code

- `<div class="row g-4">`: Bootstrap row with `g-4` gutter class adds consistent gap between columns.
- `col-sm-6`: On small screens and larger, occupies 6/12 (50%) of the row width.
- `col-lg-3`: On large screens, occupies 3/12 (25%), creating a 4-column desktop layout.
- `container`: Provides a max-width wrapper with centered margins for readable content.

- `navbar navbar-expand-lg`: Responsive navbar that expands on large screens, collapses on small.
- `d-flex align-items-center gap-2`: Flexbox utilities for display, alignment, and spacing.
- `data-bs-toggle="collapse"`: Bootstrap attribute enabling the mobile hamburger menu toggle.

Responsive Layout Behavior

- Desktop (Large Screens): Cards display 4 across (`col-lg-3`). Full horizontal navbar.
- Tablet (Medium Screens): Cards display 2 across (`col-sm-6`). Responsive navbar begins.
- Mobile (Small Screens): Cards stack vertically. Navigation collapses into hamburger menu.

Conclusion

Bootstrap is a powerful framework that enables developers to create responsive, professional websites quickly and consistently. By using Bootstrap's grid system, components, and utilities, developers eliminate the need to write basic CSS and layout code from scratch. Understanding Bootstrap is essential for modern web development.